



EXAM PAPERS PRACTICE

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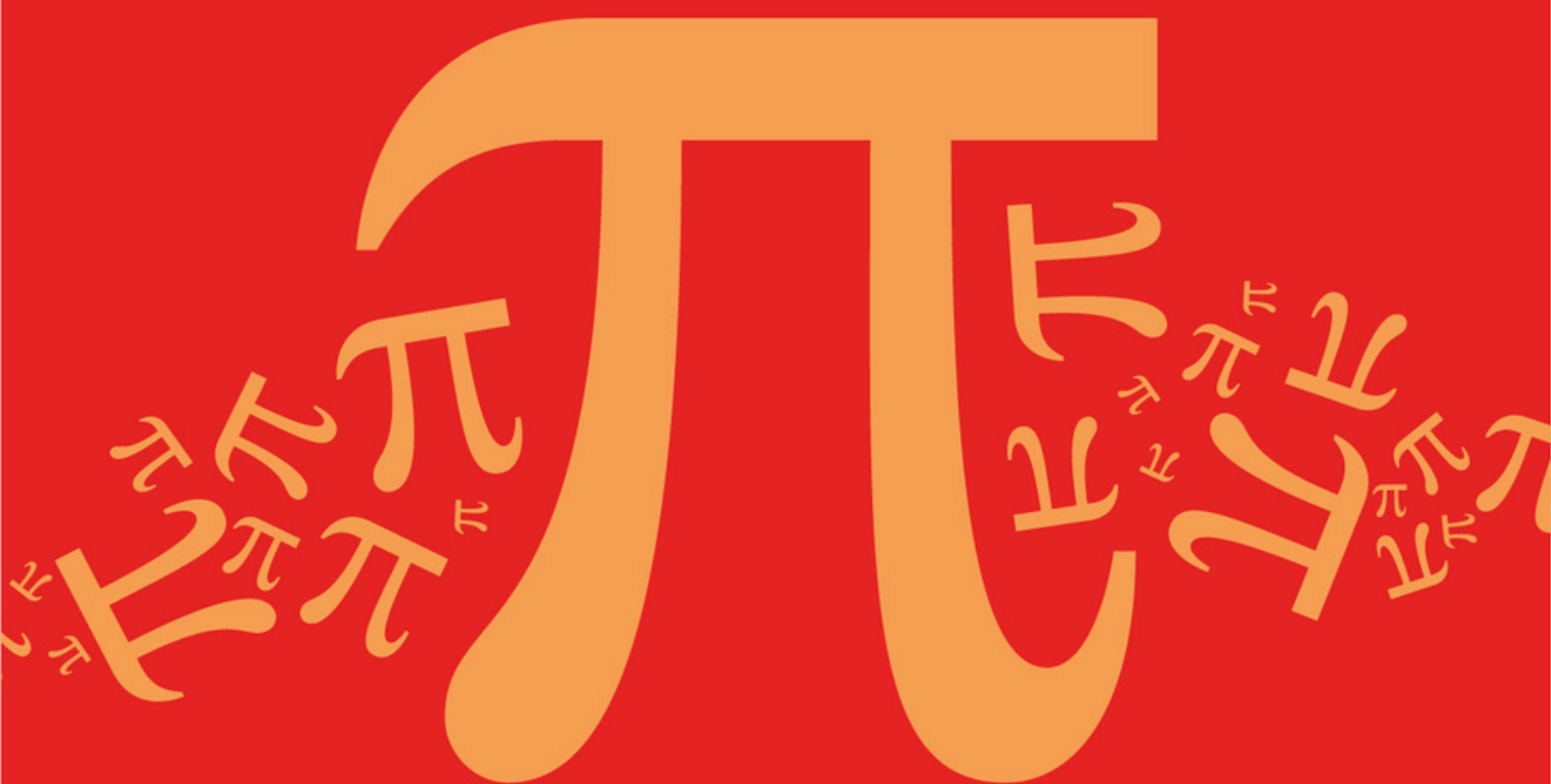
Practice questions created by actual examiners and assessment experts

Detailed mark scheme

Suitable for all boards

Designed to test your ability and thoroughly prepare you

AQA A Level Further Mathematics (7367)



Topic

C: Matrices

Sub topic

C8: Geometric Interpretation of Solutions

Mark Scheme

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Mark Scheme

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C: Matrices

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C8: Geometric Interpretation of Solutions

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**To be used by all students and teachers for
AQA A Level Further Mathematics (7367)
for exams from 2025 onwards.**

Students and teachers of other boards may also find this useful

Mark schemes

Q1.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Uses an appropriate method for finding the values of k (for example expanding appropriate determinant)	AO1.1a	M1	$\begin{vmatrix} 1 & -1 & k \\ k & -3 & 5 \\ 1 & -2 & 3 \end{vmatrix} = 0$
	Obtains a quadratic equation in k	AO1.1a	M1	$\begin{vmatrix} -3 & 5 \\ -2 & 3 \end{vmatrix} + \begin{vmatrix} k & 5 \\ 1 & 3 \end{vmatrix} + k \begin{vmatrix} k & -3 \\ 1 & -2 \end{vmatrix} = 0$
	Obtains two correct values for k	AO1.1b	A1	$1 + 3k - 5 + k(-2k + 3) = 0$ $-2k^2 + 6k - 4 = 0$ $k^2 - 3k + 2 = 0$ $(k - 2)(k - 1) = 0$ $k = 2 \text{ or } 1$
(b)	Selects an appropriate method to determine the appropriate geometrical configuration and substitutes 'their' first value of k	AO3.1a	M1	when $k = 1$ $x - y + z = 3$ $x - 3y + 5z = -1$ $x - 2y + 3z = -4$
	Eliminates one variable or uses row reduction	AO1.1a	M1	$-2y + 4z = -4$ $y - 2z = 7$
	Obtains a contradiction and makes correct deduction about the geometric configuration (must have correct value for k)	AO2.2a	R1	$y - 2z = 2; \quad y - 2z = 7$ Hence equations are inconsistent and the three planes form a prism
	Substitutes 'their' 2nd value of k into selected method to determine the appropriate geometrical configuration	AO1.1a	M1	when $k = 2$ $x - y + 2z = 3$ $2x - 3y + 5z = -1$ $x - 2y + 3z = -4$
	Obtains a consistent set of equations and makes correct deduction about geometric configuration	AO2.2a	R1	$R_2 - 2R_1 : -y + z = -7$ $R_3 - R_1 : -y + z = -7$

(must have correct value for k)			Hence equations are consistent and the three planes form a sheaf – they meet in line
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(c)	Deduces that the planes must meet in a line and hence that $k = 2$	AO2.2a	R1	$x - y + 2z = 3$ $2x - 3y + 5z = -1$ $x - 2y + 3z = -4$ $\Rightarrow -y + z = -7$ Let $z = \lambda$ Then $y = \lambda + 7$ and $x = 3 + y - 2z$ $= 3 + \lambda + 7 - 2\lambda$ $= -\lambda + 10$
	Selects method to find solution: For example, sets one variable $= \lambda$, substitutes and attempts to find other variables in terms of λ	AO1.1a	M1	ALT $\begin{pmatrix} 1 \\ -1 \\ 2 \end{pmatrix} \times \begin{pmatrix} 1 \\ -2 \\ 3 \end{pmatrix} = \begin{pmatrix} 1 \\ -1 \\ -1 \end{pmatrix}$
	Fully states correct solution CAO	AO1.1b	A1	$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 10 \\ 7 \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ -1 \\ -1 \end{pmatrix}$
Total 11 marks				

Q2.

- (a) For attempt at / getting normal d.v. $\begin{bmatrix} 1 \\ 5 \\ 7 \end{bmatrix}$

M1
A1

For getting a point on the line
eg $(0, -8, -11)$

M1
A1

For line equation: $(\mathbf{r} =)$ (their point) + λ (their d.v.)
Condone lack of r or r here

B1

ALT

eg Adding Π_1 and Π_2

(M1)

$$\Rightarrow 5x - y = 8$$

(B1)

Setting (eg) $x = \lambda$ and getting y, z in terms of λ :
 $y = 5\lambda - 8, z = 7\lambda - 11$

(M1)

Turning this into a vector equation of the given form

(M1)

$$(\mathbf{r} =) \begin{bmatrix} 0 \\ -8 \\ -11 \end{bmatrix} + \lambda \begin{bmatrix} 1 \\ 5 \\ 7 \end{bmatrix}$$

(A1)

5

- (b) (i) Substituting their x, y, z in terms of λ into Π_3 's equation
 $(12x - y - z = 40)$

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M1

For correct statement with no λ 's in: $(19 = 40)$
ft on incorrect values from their "point"

A1

Correct conclusion, from valid working, that the system is inconsistent
or "consistent" if it genuinely yields $40 = 40$

B1

3

- (ii) The three planes form a (triangular)
 prism – allow clear diagram
ft "sheaf" from a "consistent" conclusion

B1

1

- (c) (i) e.g. $\Pi_2 + \Pi_3 \Rightarrow 15x - 3y = 45$

M1

Setting $x = 0$ (eg)

M1

Any correct point, eg $(0, -15, -25)$,
 $(3, 0, -4)$, $(\frac{23}{7}, \frac{20}{7}, 0)$ $(4, 5, 3)$ etc

A1

3

(ii) $\mathbf{r} = \begin{bmatrix} 3 \\ 0 \\ -4 \end{bmatrix} + \lambda \begin{bmatrix} 1 \\ 5 \\ 7 \end{bmatrix}$

ft their common point and d.v. from (a)
Penalise missing r or $\mathbf{r}$ here

B1

1

[13]

Q3.

(a) $\begin{vmatrix} t & 2 & 3 \\ 2 & 3 & -t \\ 3 & 5 & t+1 \end{vmatrix} = 8t^2 - 7t - 1 = 0$

Attempt at det. of coefft. mtx. (or equivalent)

M1

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Equating to zero and solving a quadratic eqn. in t

M1

$$t = 1, -\frac{1}{8}$$

A1

3

(b)
$$\begin{aligned} x + 2y + 3z &= a \\ t=1 \Rightarrow 2x + 3y - z &= b \\ 3x + 5y + 2z &= c \end{aligned}$$

FT any integer value found

B1✓

E.g. $\textcircled{1} + \textcircled{2} - \textcircled{3} \Rightarrow a + b = c$

M1 A1

3

[6]

Q4.

(a)
$$\begin{vmatrix} 2 & 1 & 3 \\ 5 & -2 & a+1 \\ a & 2 & 4 \end{vmatrix} = a^2 + 3a - 10$$

Attempt at det of coeff matrix

M1

Correct (accept unsimplified)

A1

Equating to 0 and solving quadratic in a

SC: B1 for verifying $a = 2$

$a = 2, -5$

B1 for verifying $a = -5$

m1

A1

4

(b)
$$\begin{array}{rcl} 2x + y + 3z & = & 3 \\ 5x - 2y + 3z & = & 3 \\ 2x + 2y + 4z & = & b \end{array}$$

B1

Eliminations leading to two equations in two variables

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M1

Further elimination leading to value of b

m1

$b = 4$

A1

ALT: Finding two variables in terms of third

eg $y = x$ and $z = 1 - x$

(M1)

Substituting into third equation

(m1)

$b = 4$

(A1)

4

Q5.

(a)
$$\begin{vmatrix} 1 & -2 & k \\ k+1 & 3 & 0 \\ 2 & 1 & k-1 \end{vmatrix} = 3k^2 - 2k - 5$$

Attempt at det. of coefft. mtx.

M1

Correct

A1

Setting det. = 0 and solving for k

M1

$$k = \frac{5}{3}, -1$$

A1

4

(b) $5 = x - 2y + \frac{5}{3}z$

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$k = \frac{5}{3} \Rightarrow \frac{5}{3} = \frac{8}{3}x + 3y$
ft

B1

$$3 = 2x + y + \frac{2}{3}z$$

Eliminating one variable twice

M1

$$8x + 9y = 5/15y - 8z = -21 / 5x + 3z = 11$$

Correct eqn. once; twice

A1;A1

$$5 = x - 2y - z$$

$$k = -1 \Rightarrow -1 = 3y$$

ft

B1

$$3 = 2x + y - 2z$$

Eliminating one variable twice

M1

$$x - z = \frac{13}{3} / 2x - 2z = \frac{10}{3} / 5x - 5z = 11$$

Inconsistency correctly shown

A1

OR

$$y = -\frac{1}{3} \text{ and } y = -\frac{7}{5} \text{ found}$$

A1

8

[12]

Q6.

(a) **Either** $\begin{bmatrix} 5 \\ 3 \\ 4 \end{bmatrix} \times \begin{bmatrix} 1 \\ 6 \\ 2 \end{bmatrix} = -3 \begin{bmatrix} 6 \\ 2 \\ -9 \end{bmatrix}$

Correct vector product only here

M1A1

Explaining that d.v. of L is in dirn. of Π 's nml. $\Rightarrow L \perp^\Gamma \Pi$

B1

Or $\begin{bmatrix} 6 \\ 2 \\ -9 \end{bmatrix} \cdot \begin{bmatrix} 5 \\ 3 \\ 4 \end{bmatrix} = 0 \text{ and } \begin{bmatrix} 6 \\ 2 \\ -9 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 6 \\ 2 \end{bmatrix} = 0$

(M1)(A1)

Explaining that d.v. of L is $\perp^\Gamma$ to 2 (non-//) vectors in $\Pi \Rightarrow L \perp^\Gamma \Pi$

Not just stating

(B1)

3

(b) E.g. $6 \times \textcircled{1} - \textcircled{2} : 46 = 34p + 27q$

Eliminating r from any pair of eqns.

M1

$$2 \times \textcircled{1} - \textcircled{3}: -57 = 21p + 6q$$

$$\textcircled{2} - 3 \times \textcircled{3}: -217 = 29p - 9q$$

Any 2 correct eqns (1 mark each)

A1A1

$$2 \times \textcircled{4} + 9 \times \textcircled{5}: 605 = -121p$$

Solving a 2×2 system (any means) in order to get values for p, q, r

M1

$$p = -5, q = 8, r = -1$$

All 3 ✓ CAO

A1

5

(c) $7 + 6t = -2 + 5\lambda + \mu$

(i) $8 + 2t = 0 + 3\lambda + 6\mu$

$50 - 9t = -25 + 4\lambda + 2\mu$

Including re-arrangement attempt

M1

$$9 = -6t + 5\lambda + \mu$$

$$\rightarrow 8 = -2t + 3\lambda + 6\mu$$

$$75 = 9t + 4\lambda + 2\mu$$

i.e. the above system with $p = -t, q = \lambda$ and $r = \mu$

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A1

2

(ii) Substg. $t = 5$ into L 's eqn.

Or $\lambda = 8$ and $\mu = -1$ into IT 's eqn.

M1

$$P = (37, 18, 5)$$

CAO

A1

2

[12]

Q7.

(a) $x - 2y = -1 - \lambda$

$-x + y = 3 - 3\lambda$

At least one correct from setting $z = \lambda$

B1

Solving for x and y in terms of λ

M1

$$x = 7\lambda - 5 \text{ and } y = 4\lambda - 2$$

CAO

A1

3

(b) Subst^g. x, y, z in terms of λ in $5x + ky + 17z = 1$

M1

$$35\lambda - 25 + k(4\lambda - 2) + 17\lambda - 1 = 0$$

Factsn. attempt: $(4y - 2)(k + 13) = 0$

dM1

$$(2y - 1)(k + 13) = 0$$

ANSWER GIVEN

A1

3

(c) (i) When $k = -13$, $5x - 13y + 17z$
 $= 35\lambda - 25 - 52\lambda + 26 + 17\lambda \equiv 1$

Subst^g. into 3rd eqn. and demonstrating consistency

B1

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The three planes intersect in a line

B1

$$\text{Solns. } x = 7\lambda - 5, y = 4\lambda - 2, z = \lambda$$

ft

B1F

(ii) When $k \neq -13$, $\lambda = \frac{1}{2}$

B1

$$\text{Soln. } (-1\frac{1}{2}, 0, \frac{1}{2})$$

ft

B1F

Three planes meet at a point

B1

6

Q8.

(a) $\text{eg } 3 \times (1) - (2) \Rightarrow 13y + 13z = -13$
Eliminating first variable

M1

$(3) - (2) \Rightarrow 15y + 11z = -5$

A1A1

Solving 2×2 system

M1

$x = 6, y = 1\frac{1}{2}, z = -2\frac{1}{2}$

A1

5

Alt I (Cramer's Rule):

$$\Delta = \begin{vmatrix} 1 & 3 & 5 \\ 3 & -4 & 2 \\ 3 & 11 & 13 \end{vmatrix}, \Delta_x = \begin{vmatrix} -2 & 3 & 5 \\ 7 & -4 & 2 \\ 2 & 11 & 13 \end{vmatrix}$$

$$\Delta_y = \begin{vmatrix} 1 & -2 & 5 \\ 3 & 7 & 2 \\ 3 & 2 & 13 \end{vmatrix}, \Delta_z = \begin{vmatrix} 1 & 3 & -2 \\ 3 & -4 & 7 \\ 3 & 11 & 2 \end{vmatrix}$$

Attempt at any two

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= 52, 312, 78 and -130 respectively

Δ correct; ≥ 1 other determinant correct

(A1A1)

$$x = \frac{\Delta_x}{\Delta}, y = \frac{\Delta_y}{\Delta}, z = \frac{\Delta_z}{\Delta}$$

At least one attempted numerically

(M1)

$x = 6, y = 1\frac{1}{2}, z = -2\frac{1}{2}$

(A1)

(5)

Alt II (Augmented matrix method):

$$\left[\begin{array}{ccc|c} 1 & 3 & 5 & -2 \\ 3 & -4 & 2 & 7 \\ 3 & 11 & 13 & 2 \end{array} \right] \rightarrow$$

(M1)

$$\left[\begin{array}{ccc|c} 1 & 3 & 5 & -2 \\ 0 & -13 & -13 & 13 \\ 0 & 2 & -2 & 8 \end{array} \right]$$

$$\begin{aligned} R_2 &\rightarrow R_2 - 3R_1 \\ R_3 &\rightarrow R_3 - 3R_1 \end{aligned}$$

(A1)

$$\rightarrow \left[\begin{array}{ccc|c} 1 & 3 & 5 & -2 \\ 0 & 1 & 1 & -1 \\ 0 & 1 & -1 & 4 \end{array} \right]$$

(A1)

$$\rightarrow \left[\begin{array}{ccc|c} 1 & 3 & 5 & -2 \\ 0 & 1 & 1 & -1 \\ 0 & 0 & -2 & 5 \end{array} \right]$$

$$R_3 \rightarrow R_3 - R_2$$

Substituting back to get
 $x = 6, y = 1\frac{1}{2}, z = -2\frac{1}{2}$

(M1A1)

(5)

Alt III (Inverse matrix method):

$$C^{-1} = \frac{1}{52} \begin{bmatrix} -74 & 16 & 26 \\ -33 & -2 & 13 \\ 45 & -2 & -13 \end{bmatrix}$$

M0 if no inverse matrix is given

(M1A1A1)

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = C^{-1} \begin{bmatrix} -2 \\ 7 \\ 2 \end{bmatrix} = \begin{bmatrix} 6 \\ 1.5 \\ -2.5 \end{bmatrix}$$

(M1A1)

(5)

(b) (i)
$$\begin{vmatrix} 1 & 3 & 5 \\ 3 & -4 & 2 \\ a & 11 & 13 \end{vmatrix} = 26a - 26$$

Attempt at determinant; OE

M1

Setting equal to zero and solving for a

m1

$$a = 1$$

A1

3

(ii) $x + 3y + 5z = -2$
 $3x - 4y + 2z = 7$
 $x + 11y + 13z = b$

NB $y + z = -1$ (from before)

B1

$$(3) - (1) \Rightarrow 8y + 8z = b + 2$$

B1

$$b + 2 = -8 \Rightarrow b = -10$$

Equating; CAO

M1A1

4

Alternative:

Substituting $x = 6$, $y = 1\frac{1}{2}$, $z = -2\frac{1}{2}$
 into $x + 11y + 13z = b$

(M3)

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$$\Rightarrow b = -10$$

Since, to be consistent, the 3rd plane must contain the line of intersection of the first 2 planes, and therefore contains this point

(A1)

(4)

[12]

Q9.

(a) eg $(2) - (1) \Rightarrow x + 7z = -3$
 Eliminating first variable

M1A1

$$(3) - 2 \times (2) \Rightarrow x + 8z = -2$$

A1

Solving 2×2 system

M1

$$x = -10, y = 19, z = 1$$

A1

5

(b) (i)

$$\begin{vmatrix} 1 & 1 & -3 \\ 2 & 1 & 4 \\ 5 & 2 & a \end{vmatrix} = 15 - a$$

Determinant

B1

Setting = to zero and solving for a
Must get a numerical answer

M1

$$a = 15$$

ft

A1

3

(ii)

$$\begin{aligned} x + y - 3z &= b \\ 2x + y + 4z &= 3 \\ 5x + 2y + 15z &= 4 \\ \text{eg } (2) - (1) &\Rightarrow x + 7z = 3 - b \end{aligned}$$

Eliminating first variable

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M1A1

$$(3) - 2 \times (2) \Rightarrow x + 7z = -2$$

A1

Equating the two RHSs

M1

$$b = 5$$

CAO

NB Eliminating x : $-y + 10z = 3 - 2b$

$$-3y + 30z = 4 - 5b$$

$$-y + 10z = -7$$

NB Eliminating z : $10x + 7y = 4b + 9$

$$10x + 7y = 5b + 4$$

$$10x + 7y = 29$$

A1

5

Alternate Schemes

(a) Cramer's Rule

$$\Delta = \begin{vmatrix} 1 & 1 & -3 \\ 2 & 1 & 4 \\ 5 & 2 & 16 \end{vmatrix}, \quad \Delta_x = \begin{vmatrix} 6 & 1 & -3 \\ 3 & 1 & 4 \\ 4 & 2 & 16 \end{vmatrix},$$

$$\Delta_y = \begin{vmatrix} 1 & 6 & -3 \\ 2 & 3 & 4 \\ 5 & 4 & 16 \end{vmatrix}, \quad \Delta_z = \begin{vmatrix} 1 & 1 & 6 \\ 2 & 1 & 3 \\ 5 & 2 & 4 \end{vmatrix}$$

Attempt at any two

= -1, 10, -19 and -1 respectively

Any one correct

$$x = \frac{\Delta_x}{\Delta}, \quad y = \frac{\Delta_y}{\Delta}, \quad z = \frac{\Delta_z}{\Delta}$$

At least one attempted numerically

$x = -10, y = 19, z = 1$

Any 2 correct ft

All 3 correct CAO

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M1

A1

M1

A1

A1

(5)

(a) Inverse matrix method

$$C^{-1} = \frac{1}{-1} \begin{bmatrix} 8 & -22 & 7 \\ -12 & 31 & -10 \\ -1 & 3 & -1 \end{bmatrix}$$

M0 if no inverse matrix is given

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = C^{-1} \begin{bmatrix} 6 \\ 3 \\ 4 \end{bmatrix} = \begin{bmatrix} -10 \\ 19 \\ 1 \end{bmatrix}$$

Any 2 correct ft

All 3 correct CAO

M1A1

M1

A1

A1

(5)

Alternate scheme

$$(all) \quad \left[\begin{array}{ccc|c} 1 & 1 & -3 & b \\ 2 & 1 & 4 & 3 \\ 5 & 2 & a & 4 \end{array} \right] \rightarrow \left[\begin{array}{ccc|c} 1 & 1 & -3 & b \\ 0 & -1 & 10 & 3-2b \\ 0 & -3 & a+15 & 4-5b \end{array} \right]$$

$$R_2' = R_2 - 2R_1$$

$$R_3' = R_3 - 5R_1$$

$$\rightarrow \left[\begin{array}{ccc|c} 1 & 1 & -3 & b \\ 0 & 1 & -10 & 2b-3 \\ 0 & 0 & a-15 & b-5 \end{array} \right]$$

$$R_3' = R_3 + 3R_2$$

(4)

(b) (i) For non-unique solutions, $a = 15$

(2)

(ii) For consistency,
 $4 - 5b = 3(3 - 2b) \Rightarrow b = 5$

(2)

(a) When $a = 16$, $b = 6$

$$\left[\begin{array}{ccc|c} 1 & 1 & -3 & 6 \\ 0 & 1 & -10 & 9 \\ 0 & 0 & 1 & 1 \end{array} \right]$$

$$\Rightarrow z = 1, y = 19, x = -10$$

(5)

[13]

Q10.

$$\begin{array}{ccc|ccc|c} 1 & 2 & -1 & 0 & 1 & 2 & -1 & 0 \\ 3 & -1 & 4 & 7 & 0 & -1 & 7 & 7 \\ 8 & 1 & 7 & 30 & 0 & -15 & 15 & 30 \end{array} \rightarrow \begin{array}{ccc|c} 1 & 2 & -1 & 0 \\ 0 & -1 & 7 & 7 \\ 0 & -15 & 15 & 30 \end{array}$$

$$R_2' = R_2 - 3R_1$$

M1

$$R_3' = R_3 - 8R_1$$

Penalise numerical errors once only, at this stage

A1

$$\begin{array}{ccc|c} 1 & 2 & -1 & 0 \\ \rightarrow 0 & -1 & 1 & 1 \\ 0 & -1 & 1 & 2 \end{array}$$

A1

*Inconsistency noted/explained
ft provided working is clear*

B1F

4

Or $\Delta = -7 - 3 + 64 - 8 - 4 - 42 = 0$

(M1)
(A1)

and Δ_x or Δ_y or $\Delta_z = 0$ shown also

(A1)

Explaining this $\Rightarrow$ inconsistency

*So showing $\Delta = 0$ and thinking this is it
scores M1A1A0B0*

(B1)

(4)

Or Solving (1) & (2), say, to get

(M1)

$x = \lambda, y = 1 - \lambda, z = 2 - \lambda$

(A1)
(A1)

Substⁿ. in (3) $\Rightarrow 15 = 30$

Checking to show inconsistency

(B1)

(4)

[4]

Q11.

(a) $\Delta = \begin{vmatrix} k & 2 & 1 \\ 1 & k+1 & -2 \\ 2 & -k & 3 \end{vmatrix}$

$= 3k^2 + 3k - k - 8 - 2(k+1) - 2k^2 - 6$

Genuine attempt at Δ

M1

$$= k^2 - 16$$

A1

When $k^2 = 16 \Delta = 0 \Rightarrow$ no unique soln.

Explained

E1

3

Or Subst⁹. **Both** $k = 4$ and $k = -4$ and attempt at det

(M1)

Each case correctly shown

(A1)

(A1)

3

(b)

$$4x + 2y + z = 5$$

$$k = 4 \Rightarrow x + 5y - 2z = 3$$

$$2x - 4y + 3z = -11$$

B1

Elim⁹. z from **(1) & (2)** $\Rightarrow 9(x + y) = 13$

Eliminating one variable

M1

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$$\mathbf{(1) \& (3)} \Rightarrow 10(x + y) = 26$$

Twice, correctly

A1

$$\text{Or } \mathbf{(2) \& (3)} \Rightarrow 7(x + y) = -13$$

Explaining inconsistency, eg from $\frac{13}{9} \neq \frac{26}{10}$

E1

4

Alternatively (mark as above)

Elim⁹. x from **(1) & (2)** $\Rightarrow 9(2y - z) = 7$

$$\mathbf{(2) \& (3)} \Rightarrow 7(2y - z) = 17$$

$$\mathbf{(1) \& (3)} \Rightarrow 5(2y - z) = 27$$

Or

Elim⁹. y from **(1) & (2)** $\Rightarrow 9(2x + z) = 19$

$$\mathbf{(2) \& (3)} \Rightarrow 7(2x + z) = -43$$

$$(1) \text{ \& } (3) \Rightarrow 5(2x + z) = -1$$

(c) (i) $-4x + 2y + z = 5$

$$k = -4 \Rightarrow x - 3y - 2z = 3$$

$$2x + 4y + 3z = -11$$

B1

Eliminating one variable
Any pair of equations

M1

$$-7x + y = 13$$

$$\text{Or } 10y + 7z = -17$$

$$\text{Or } 10x + z = -21$$

Correct

A1

Parametrisation
Or equivalent

M1

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 0 \\ 13 \\ -21 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 7 \\ -10 \end{pmatrix}$$

Any correct answer in any form

A1

5

Correct alternate answer forms:

$$x, y = 13 + 7x, z = -21 - 10x$$

$$y, x = (y - 13) / 7, z = (-21 - 10y) / 7$$

$$z, y = (-17 - 7z) / 10, x = (-21 - z) / 10$$

Do not accept a mixed parametrisation

Or equivalents

(ii) The line of intersection of 3 planes
Or "Sheaf" of planes

B1

1

[13]

Q12.

(a)
$$\Delta = \begin{vmatrix} 0 & 0 & 1 \\ a-c & b-c & c \\ b(c-a) & a(c-b) & ab \end{vmatrix}$$

By $C_1' = C_1 - C_3$ and $C_2' = C_2 - C_3$

M1

$$(a-c)(b-c) \begin{vmatrix} 0 & 0 & 1 \\ 1 & 1 & c \\ -b & -a & ab \end{vmatrix}$$

A1

1st and 2nd factors

A1

$$= (a-c)(b-c)(b-a)$$

By expanding

M1

$$= (a-b)(b-c)(c-a)$$

AG

A1

5

(b) (i) Noting $a = b = 3, c = 5$ in coefft. matrix

M1

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$$\Rightarrow \Delta = 0 \text{ since factor } (a-b) = 0$$

Or starting from scratch

Hence no unique soln. to system

Clearly explained

A1

2

(ii) $3R_2 + R_3 = 24R_1$

M1

giving consistency iff $24p - 3q - r = 0$

Answer given

A1

2

(iii) $x + y + z = 1$

$$3x + 3y + 5z = 8$$

$$5x + 5y + 3z = 0$$

$$(2) - 3 \times (1) \text{ or } (3) - 5 \times (1) \Rightarrow z = 2\frac{1}{2}$$

M1A1

$$\text{Then } x + y = -1\frac{1}{2}$$

B1

$$\text{Setting } x = \lambda (\text{e.g.}) \Rightarrow y = -1\frac{1}{2} - \lambda$$

Parametrisation

M1

All correct, any form
CAO

A1

5

Special Case ruling for those who don't attempt to parametrise
but who show this *is* the full solution:
4 + B0

First alternative (a)

$$\text{Setting } a = b \text{ (etc.)} \Rightarrow \Delta = 0 \text{ since } C_1 = C_2$$

M1

$$\Rightarrow (a - b) \text{ a factor, by the factor theorem}$$

A1

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By cyclic symmetry, $(b - c)$ and $(c - a)$ are also

M1

factors

A1

Checking sign/coefft, = (+) 1

A1

5

Second alternative (a)

Expanding the determinant

M1

$$\Delta = ab^2 + a^2c + bc^2 - b^2c - ac^2 - a^2b$$

A1

Multiplying out $(a - b)(b - c)(c - a)$

No fudging, or jumping straight to the answer allowed

M1

$$= abc + ab^2 + a^2c + bc^2 - b^2c - ac^2 - a^2b - abc$$

A1

Fully correct working to show the 2 things =

A1

5

[14]

Q13.

(a)
$$\begin{vmatrix} 1 & 3 & -1 \\ 2 & k & 1 \\ 3 & 5 & k-2 \end{vmatrix}$$

Attempted expansion

M1

$$= k^2 - 2k - 10 + 9 + 3k - 5 - 6(k - 2) = k^2 - 5k + 6$$

AG

A1

2

(b) (i) When $k = 1$, $\Delta \neq 0 \Rightarrow \exists$ **one** solution.

Give for correct single soln.

(-16, 13.5, 14.5) found

B1

(ii) When $k = 2$, $\Delta = 0$ (no unique solution.)

System is $x + 3y - z = 10$

$$2x + 2y + z = -4$$

$$3x + 5y = 6$$

B1

$R_1 + R_2 = R_3$ on *both* sides

M1

$\Rightarrow \exists$ ∞ -ly many solutions

A1

(iii) When $k = 3$, $\Delta = 0$ (no unique solution.)

System is $x + 3y - z = 10$

$$\begin{aligned} 2x + 3y + z &= -4 \\ 3x + 5y + z &= 7 \end{aligned}$$

B1

$$\begin{aligned} (1) + (2) &\Rightarrow 3x + 6y = 6 \\ \text{Or } x + 2y &= 2 \end{aligned}$$

$$\begin{aligned} (1) + (3) &\Rightarrow 4x + 8y = 17 \\ \text{Or } x + 2y &= 4\frac{1}{4} \text{ etc.} \end{aligned}$$

M1

$\Rightarrow$ System inconsistent and $\exists$ no solutions

A1

7

(c) $k = 1$: the (single) point of intersection of 3 planes

B1 ✓

$k = 2$: 3 planes meet in a line (or form a sheaf)

B1 ✓

$k = 3$: 3 planes form a “prism” (or have three parallel lines of intersection; or have no common intersection)

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B1 ✓

ft one of each type (provided matched up correctly)

3

[12]

Q14.

$$\begin{aligned} \text{(a)} \quad \Delta &= \begin{vmatrix} a+b+c & a+b+c & a+b+c \\ b+c & c+a & a+b \\ b-c & c-a & a-b \end{vmatrix} \\ &\text{By } R_1' = R_1 + R_2 \end{aligned}$$

M1A1

$$\begin{aligned} &= (a + b + c) \begin{vmatrix} 1 & 1 & 1 \\ b+c & c+a & a+b \\ b-c & c-a & a-b \end{vmatrix} \\ &\text{Factoring out correctly} \end{aligned}$$

A1

$$\text{or } \begin{vmatrix} a+b+c & b & c \\ 2(a+b+c) & c+a & a+b \\ 0 & c-a & a-b \end{vmatrix}$$

By $C_1' = C_1 + C_2 + C_3$

$$= (a+b+c) \begin{vmatrix} 1 & b & c \\ 2 & c+a & a+b \\ 0 & c-a & a-b \end{vmatrix}$$

Etc.

Method for obtaining remaining factor

$$\Delta = 2(a+b+c)(a^2 + b^2 + c^2 - ab - bc - ca)$$

2 can be anywhere

M1

A1

5

(b) Nothing $a = a, b = 3, c = 1$

B1

Setting $\Delta = 0$

M1

$$\text{i.e. } 0 = 2(a+4)(a^2 - 4a + 7) \Rightarrow a = -4$$

A1

Showing $(a-2)^2 + 3 \neq 0$ so only one value of a

Ignore incorrect quadratic factors until here

CSO

M1

A1

5

Alt.

$$\begin{vmatrix} a & 3 & 1 \\ 4 & 1+a & a+3 \\ 2 & 1-a & a-3 \end{vmatrix} = 2a^3 - 18a + 56$$

(B1)

Equating to zero + solving attempt

(M1)

$$2(a + 4)(a^2 - 4a + 7) = 0 \Rightarrow a = -4$$

(A1)

$$(a - 2)^2 + 3 > 0 \Rightarrow \text{no other real solutions.}$$

Or discriminant < 0

(M1A1)

(5)

[10]



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Examiner reports

Q2.

Responses to this question were usually much more confident and, apart from the usual batch of numerical and / or sign slips, decently correct. Finding points and directions proved to be much more what students were expecting.

Q3.

This was generally well done, with marks relatively high. It was surprising to find that those who failed to get any integer value for t generally didn't appear to go back and check their working. Weaker candidates often didn't seem to know how to go about establishing consistency.

Q4.

In part (a), most candidates realised that the determinant of the coefficients on the left-hand sides of the equations must be equated to zero, and thus obtained the required values efficiently. Solutions to part (b) were often marred by a failure to indicate which equations were being used at each step. Some candidates used the slightly dubious technique of replacing one of the variables by zero, while others produced a whole page of work and still failed to reach a value for b .

Q5.

This question was surprisingly well attempted, as the algebra that goes with the systems of equations work is usually found tough enough to guarantee lots of mistakes. Almost all candidates found the determinant of the coefficient matrix in part (a) and solved the resulting quadratic equation with ease. Other approaches generally got nowhere. The work in part (b) was also competently attempted by the majority, apart from silly arithmetical slips which again probably would have been noticed had a very quick check been made.

Q6.

There were several easy marks to be had in this question, and the rest was helpfully sign-posted. Part (a) was routine ... or it would have been had not, again, some sort of explanation been required for full marks. Approaches were evenly split between using the scalar-product and the vector-product. In the former case, a lot of candidates seemed to think they only needed to show that the line's direction vector was perpendicular to **one** vector parallel to the plane. In the second case, many candidates were content merely to find the vector product of the two vectors "in" (parallel to) the plane; others noted that it was a multiple of the d.v. of L without further comment. For the 3rd mark on this part of the question, we wanted an explanation as to *why* their scalar- or vector-product work established the given outcome – it is not enough for candidates merely to repeat the result that we gave them. *Given* results always require more of candidates by way of 'dotting the i's and crossing the t's'.

The more discerning teachers and students of MFP4 will understand why part (b) was set up as it was, since this is the only way to force candidates not just to resort to calculator-output generated answers. In part (c), remarkably few got past equating the equations of L and II ; the majority of candidates had no real idea what to do and so there were lots of answers seen in which the point of intersection was taken to be $(-5, 8, -1)$.

Q7.

This question was definitely the most demanding on the paper; partly because this topic of solving systems of equations has generally proved to be one of the least palatable topics to candidates over the years, but mostly due to the poor examination technique employed on this occasion. Many candidates ignored the directly-stated demand to set $z = \lambda$ in part (a).

It was unfortunate that few candidates seemed to notice that the given result in part (b) should be used in part (c). This resulted in much unnecessary work for many candidates and often a failure to answer the question asked. The demand was for candidates to “determine” any solutions, and then to describe what they were in relation to the planes. Many candidates ignored the first of these requests altogether, and then went on to describe the arrangement of the planes, which was not what the question said. A large majority of candidates lost four or five marks, in part (c)(i) by failing to re-state their answer to part (a) and then calling it a “line” (of intersection) and in part (c)(ii) by failing to notice the consequence of part (b)’s answer that $z = \lambda = \frac{1}{2}$, which then gives a point (of intersection) whose coordinates can be written straight down (again, using part (a)’s answer).

Q8.

It was very pleasing to see a large numbers of candidates solving part (a)’s system of equations using some fairly high-powered techniques (see the end of the marking scheme for these: Cramer’s Rule, the augmented matrix method, and the inverse matrix method) to find the unique solution, but most of these are actually more complicated than what was required. The “low-tech” algebraic approach does the job so easily that the others (whilst eminently worth teaching) often turn out to be longer and more prone to errors.

One approach that did turn up for part (b)(ii) is that one can simply substitute the values of x , y and z found in part (a) into the 3rd equation – using the value of a from part (b)(i) – in order to find the value of b required for the final equation to be consistent. This is because the third plane must share the line of intersection of the first two planes, and hence the point on it found in part (a). Only a handful of students actually employed this approach.

Q9.

Despite a pleasing array of higher-powered techniques on display in solutions here, the low-tech method of first reducing the 3×3 system to a 2×2 system is by far the easiest approach, and the one most commonly used by candidates. However, although eliminating the y -variable first makes for the least demanding working, candidates’ efforts were fairly evenly distributed amongst the three variables. Though this makes no difference in principle, the tendency for weaker candidates not to stop and think about their approach often led to unnecessarily tricky algebra later on in part (b), which then provided more opportunities for errors to occur in the solutions presented.

Q10.

Although intended as a straightforward starter, this question caused as much difficulty as any on the paper. Around a third of candidates thought that it was sufficient to show that the determinant of the coefficient matrix was zero – ie for non-uniqueness of solution(s) – while many others relied on faulty arithmetic to support their claim of inconsistency.

Q11.

This algebra question was the one that produced the most variable set of responses from

candidates, from the outstanding to the careless to the “haven’t got a clue what to do” variety. In part (a), the general approach is to find the determinant of the coefficient matrix in terms of k and see when this is zero. The alternative is to evaluate it using $k = 4$ and -4 . It was really shocking to see how few candidates appeared to realise that $k^2 = 16$ gave rise to the two values of k . As mentioned earlier, the manipulation of some fairly simple equations in order to eliminate one or other of the three variables left a lot to be desired, and many candidates would have scored several more marks with just a little more care.

Moreover, the question itself implies that there are no solutions to part (b) and infinitely many in part (c), so it was surprising to see so many submissions moving towards a unique solution to the system in these cases. Finally, in part (b), the widespread inability of candidates to demonstrate an inconsistency was disappointing. Very few candidates did so successfully, predominantly because much of their prior working was error-strewn, rendering a valid conclusion impossible, despite their claims.

Q12.

This question was a good source of marks for most candidates, although few managed to achieve full marks on it. Only around 50% of candidates knew how to manipulate determinants using row and/or column operations, although many got by on a much simpler level by expanding the determinant out to begin with, multiplying out the three brackets $(a - b)(b - c)(c - a)$, and showing that they gave the same expression. This is a valid and acceptable approach, even though not the intended one. Even so, many thought they could get away with expanding the determinant and then just saying that this was equal to the given answer $(a - b)(b - c)(c - a)$. They got two of the five marks available here. For future reference, it should be noted that handling determinants of higher order than a cubic is very unlikely to be done successfully by candidates using alternate approaches such as those described here and that straightforward row and column operations will be required to extract factors.

Answers to parts (b)(i) and (ii) were often poorly explained, despite being relatively straightforward in principle. A pleasing number of candidates realised that they could use the answer to part (a) in part (b)(i). Three of the five marks for part (b)(iii) were straightforward and most candidates got as far as $z = 2.5$ and $x + y = -1.5$.

Disappointingly, an overwhelming majority stopped here, failing to write the solutions in any obviously acceptable parameterised or vector line form, such as $(x, y, z) = (t, -1.5 - t, 2.5)$ or equivalent. Many, implicitly or explicitly, suggested that $x + y = -1.5$ is the equation of the line solution. Any candidates who explained why the answer $z = 2.5$ and $x + y = -1.5$ will suffice would, of course, have scored full marks, but satisfactory explanations were very few and far between.

Q13.

This was the worst answered question on the paper. Although seven marks were allocated to part (b), many candidates simply wrote down answers without justification. Few seemed to have a correct notion of the connection between the determinant found in part (a) and the number of solutions of the system. A curious mistake was to find values for x , y and z and then to conclude that there were three solutions to the system of equations. A small number of candidates, even though not answering part (b) correctly, were aware of the three possible answers and gave one of each. It was then possible for them to deduce the correct geometrical interpretations for each case and obtain the marks for part (c).

Q14.

This question was less well done than Question 7 for the most part. Many candidates knew how to expand a determinant, but were unable to manipulate one using row/column operations. With part (a) proving to be difficult, part (b) was usually even more difficult. Candidates often failed to use part (a) to answer part (b). Very few candidates produced a second quadratic factor for the determinant and only a minority of these justified that the required value of a was unique.



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